Muhammad Farhan

AI Engineer — LLMs & Agentic Systems · Generative AI · Python · FastAPI

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PROFESSIONAL SUMMARY

AI Engineer specialising in LLMs and Agentic AI systems, with 1+ year of hands-on experience and 10+ projects built — designing multi-agent architectures (LangChain, LangGraph) with real orchestration and state handoff, and serving them behind production-style FastAPI backends with Postgres, Redis, and Celery. IBM-certified in Generative AI Engineering and Top 5 of 35 at the GIKI Capstone Competition. Grounded in classical ML, deep learning, and reinforcement learning beneath the agentic work.

TECHNICAL SKILLS

Generative AI / Agentic	LLM Integration · Multi-Agent Systems (LangChain, LangGraph) · Prompt Engineering · RAG & Hybrid Search · Model Context Protocol (MCP) · Orchestration & State Handoff · Human-in-the-Loop · Evaluation / Guardrail Design
LLM Providers & Infra	OpenAI · Google Gemini · Anthropic Claude · FastAPI
Data & Backend	PostgreSQL (SQLAlchemy 2.0, Alembic) · ChromaDB · REST API Design · Streamlit · Flask
Deep Learning & ML	CNNs · NLP · Reinforcement Learning (PPO) · PyTorch · TensorFlow / Keras · Scikit-learn · Feature Engineering · Model Evaluation · Pandas · NumPy · Matplotlib · Seaborn
Web Development	Next.js · React · TypeScript · JavaScript · Tailwind CSS
Tools & Infra	Python (Advanced) · Git & GitHub · Docker / Docker Compose · pytest · MLflow · Jupyter
Languages	English (Professional) · Urdu (Native) · Pashto (Native)

PROFESSIONAL EXPERIENCE

AI Engineer · Capstone Programme, GIKI — 2026 – Present

Ghulam Ishaq Khan Institute of Engineering Sciences & Technology, Pakistan · Hybrid

- Building MootCourtSimulator, an AI-powered moot court platform where law students argue cases in real time against an opposing-counsel LLM agent.
- Developing an autonomous AI judge agent that scores argument quality, legal reasoning, and rebuttal strength with structured feedback.
- Architecting the multi-agent conversation flow and evaluation rubric that keep debate turns contextual and consistently graded — project selected among the Top 5 of 35 (Honorable Mention).

FEATURED PROJECTS — LLM & AGENTIC SYSTEMS

MootCourtSimulator — AI Moot Court Platform — Capstone, GIKI

Recorded Demo · GitHub

Stack: Python · TypeScript · React/Vite · LangGraph · FastAPI · Express/Drizzle ORM · PostgreSQL · OpenAI (Whisper, GPT-4o, TTS) · Docker

- Building an AI courtroom where a law student argues a live case against an AI Opposing Counsel while an AI Judge scores the reasoning against a fixed rubric (legal reasoning, argument strength, evidence use, rebuttal handling), returning a performance score with detailed feedback.
- Covers the full moot court cycle — case briefing, oral arguments, judicial questioning, rebuttals, and final judgment — via real-time voice streaming (Whisper transcription → GPT-4o reasoning → multi-voice TTS) on a 3-tier system (React/Vite client, Express/Drizzle API gateway, FastAPI/LangGraph AI service).
- Designed a Hybrid RAG pipeline over a verified 53-provision Pakistani statutory corpus (Constitution 1973, PPC 1860, CrPC 1898, QSO 1984) — dense vector search fused with BM25 via Reciprocal Rank Fusion and LLM reranking, reaching 1.00 Hit@1 on legal query benchmarks — plus deterministic citation auditing that flags fabricated provisions in real time.
- Built a reproducible LLMOps evaluation suite (MLflow-tracked) — 1.00 objection-decision recall, a 0/9 witness-fabrication rate, and 0/36 successful red-team prompt-injection attacks; containerised the full stack with Docker Compose.

AI Content Factory — Multi-Agent Content Generation & Evaluation System — FYP (Team of 2)

Live Demo · GitHub

Stack: Python · LangGraph · LangChain · FastAPI · Next.js · GPT-5-mini · Gemini 2.5 Flash · ChromaDB · Tavily · DeepEval

- Built an orchestrator agent that researches a single topic from authoritative web sources and produces a structured outline; a blog agent then drafts the full article grounded in that research, not free-generated from the prompt.

- Gated every draft behind an evaluation agent scoring factual accuracy, source reliability, structure, readability, originality, and hallucination — failing drafts loop back for regeneration until they clear the threshold (G-Eval + DeepEval, with a non-LLM citation verifier).
- Implemented LangGraph state handoff with SQLite/Postgres checkpointing (crashed runs resume without repeating LLM calls), a Human-in-the-Loop outline editor, and parallel section writing via Send() fan-out — verified with a reproducible ablation experiment on repetition rates.
- Attached generated imagery to approved articles and repurposed each into a YouTube Short, podcast, video script, X post, and LinkedIn post.

Dev Signal — AI-Powered SaaS Market-Research & PRD Generator — Full-Stack AI Platform

GitHub

Stack: Next.js 16 · TypeScript · FastAPI · Gemini (2-pass) · PostgreSQL (SQLAlchemy 2.0, Alembic) · Celery · Redis · Docker Compose

- Built an automated discovery pipeline — async HTTPX scrapers pull top discussions across r/webdev, r/SaaS, r/programming, and the Stack Overflow API.
- Designed a two-pass Gemini pipeline — Pass 1 extracts concrete pain points with severity, sentiment, and frequency; Pass 2 turns validated ones into 3–5 targeted SaaS product ideas per run, each with a full PRD (overview, problem statement, success metrics, tiered features, tech architecture, user stories, competitor analysis).
- Modelled the domain in SQLAlchemy 2.0 with UUID keys and Alembic migrations; served it behind FastAPI with Celery and Redis for long scrape jobs, and a Next.js 16 dashboard with client-side fallbacks so the UI stays up when the backend is unreachable.

DEEP LEARNING & REINFORCEMENT LEARNING

Stick Fighter — Live In-Browser Reinforcement Learning Game

Live Demo

Stack: TypeScript · HTML5 Canvas · Vite · PPO · PyTorch · Gymnasium

- Built a 2D fighting game whose opponent isn't scripted or pre-trained — it runs PPO live in the browser, learning from the player mid-fight; hand-rolled the network (manual backprop, Adam) with no TensorFlow.js, ONNX, or WASM, in a 51 KB bundle (17 KB gzipped).
- Implemented real PPO on an actor-critic MLP (46-dim observation → 2×64 tanh trunk → 15-action policy head + value head) with GAE(λ), clipped surrogate objective, entropy bonus, grad-norm clipping, and target-KL early stopping, updating every 128 frames, with a decaying scripted prior to solve cold start.
- Debugged a PPO correctness bug where entropy stayed pinned at the uniform ceiling (ln 15) — the fix moved value loss 287 → 0.45 and approx-KL 0.46 → 0.003; also found and fixed a frame-data exploit in the fight engine.
- With the scripted prior stripped, the learned agent won 12/12 rounds (vs. 11/12) in 258 frames (vs. 619) with 91 HP remaining (vs. 43), verified by a 43-check headless test suite.

OTHER ML PROJECTS

- Customer Churn Prediction (Scikit-learn) · URL Phishing Detection · House Price Regression — models with full evaluation pipelines covering feature engineering, preprocessing, and metrics (accuracy, precision, recall, ROC-AUC, RMSE/R^2).

EDUCATION

BS Computer Science · University of Peshawar, Pakistan — 2022 – 2026

- All 8 semesters completed; degree awaiting official conferral. Available for full-time roles immediately.
- Relevant coursework: Machine Learning · Deep Learning · Data Structures & Algorithms · Databases · Software Engineering.
- Final Year Project: AI Content Factory — multi-agent blog generation system using LLM agent orchestration (LangGraph).

CERTIFICATIONS & TRAINING

- IBM Generative AI Engineering Professional Certificate (16 courses) — IBM · Coursera · Sep 2026 Verify Certificate
- IBM Introduction to Computer Vision and Image Processing — IBM · Coursera · Sep 2026 Verify Certificate
- Top 5 Honorable Mention — GIKI Capstone Competition (MootCourtSimulator), Asher Aziz Foundation · Aug 2026 View Certificate
- Advanced AI Bootcamp (Grade B+) — GIK Institute & Asher Aziz Foundation · Aug 2026 View Certificate
- MERN Stack Development (Grade A+) — NAVTTC · Prime Minister's Youth Skills Development Program View Certificate
- Web and App Development — Saylani Mass IT Training (SMIT) View Certificate
- Fundamentals of Deep Learning · Fundamentals of Machine Learning — NVIDIA (Coursera)